Automated COVID-19 Data Collection (2022–2024) Using Python and API Integration

Project Overview

This project automates the process of collecting and cleaning daily COVID-19 data for 15 countries between January 2022 and December 2024. Data was fetched directly from the Our World in Data (OWID) COVID-19 dataset using Python. The cleaned and structured dataset provides a reliable source for further visualization or analysis.

Objectives

- Collect daily COVID-19 data for 15 countries (2022–2024)
- Include New Cases, New Deaths, Total Vaccinations, Stringency Index
- Handle missing values and inconsistencies
- Export final dataset to CSV

Tools & Technologies

- Python (pandas, requests)
- Jupyter Notebook / VS Code
- Data Source: Our World in Data (OWID COVID-19 dataset)
- Output Format: CSV

Data Pipeline

1. Retrieve dataset via OWID API
2. Filter for 15 selected countries
3. Clean data and handle missing values
4. Format date range (2022–2024)
5. Export results to CSV

Dataset Columns

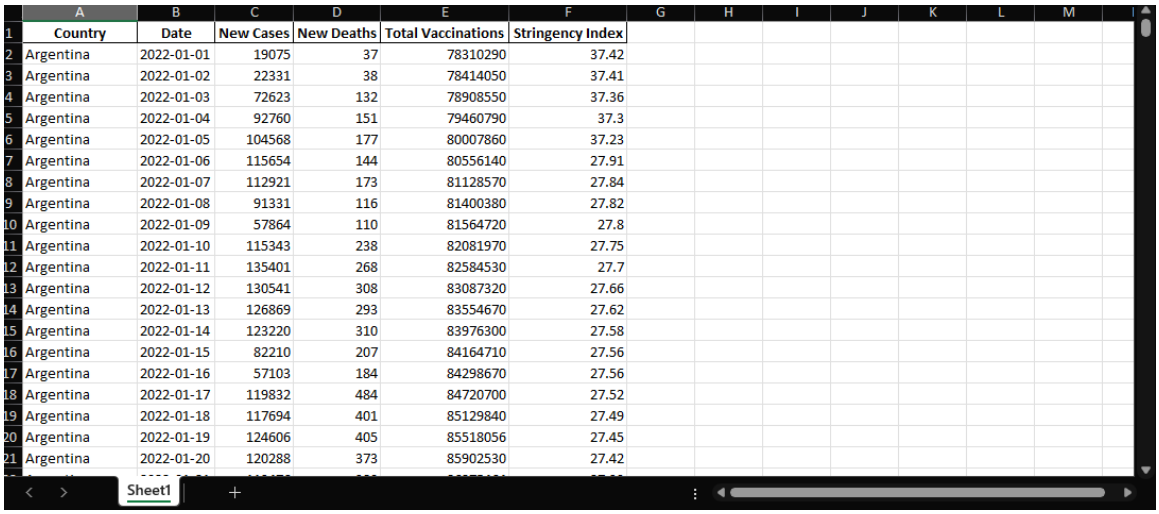
Column Name	Description
Country	Country name
Date	Date (YYYY-MM-DD)
New Cases	Daily new cases
New Deaths	Daily new deaths
Total Vaccinations	Cumulative vaccinations
Stringency Index	Government response index

Deliverables

- covid_data_collector.py (Python script)
- covid_2022_2024_15countries.csv (clean dataset)
- README.md (documentation)
- This report (for GitHub/Upwork)

Conclusion

This project demonstrates an automated data collection pipeline using Python for real-world COVID-19 data. The resulting dataset is clean, consistent, and ready for analysis or visualization.



Country	Date	New Cases	New Deaths	Total Vaccinations	Stringency Index
Argentina	2022-01-01	19075	37	78310290	37.42
Argentina	2022-01-02	22331	38	78414050	37.41
Argentina	2022-01-03	72623	132	78908550	37.36
Argentina	2022-01-04	92760	151	79460790	37.3
Argentina	2022-01-05	104568	177	80007860	37.23
Argentina	2022-01-06	115654	144	80556140	27.91
Argentina	2022-01-07	112921	173	81128570	27.84
Argentina	2022-01-08	91331	116	81400380	27.82
Argentina	2022-01-09	57864	110	81564720	27.8
Argentina	2022-01-10	115343	238	82081970	27.75
Argentina	2022-01-11	135401	268	82584530	27.7
Argentina	2022-01-12	130541	308	83087320	27.66
Argentina	2022-01-13	126869	293	83554670	27.62
Argentina	2022-01-14	123220	310	83976300	27.58
Argentina	2022-01-15	82210	207	84164710	27.56
Argentina	2022-01-16	57103	184	84298670	27.56
Argentina	2022-01-17	119832	484	84720700	27.52
Argentina	2022-01-18	117694	401	85129840	27.49
Argentina	2022-01-19	124606	405	85518056	27.45
Argentina	2022-01-20	120288	373	85902530	27.42

Sheet1